Pedestrian Fatalities: Causal Discovery

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Abstract

We use the Really Fast Causal Inference (RFCI) algorithm (Colombo et al., 2012) to estimate a Partial Ancestral Graph (PAG) based on conditional independence tests. We refine this PAG with domain knowledge to infer undirected edges for formulating a hypothesized Directed Acyclic Graph (DAG). This method is applied to 250,000 fatal crashes records from National Highway Traffic Safety Administration data between 2012 and 2024. Our analysis provides evidence for several causal claims among fatal crashes. The data suggest that weather mainly affects pedestrian death through lighting conditions, previous driver accidents are not related to pedestrian fatalities, and some of age's effects are directed through interesting mediators. Our method identifies the following six variables mediating all effects on pedestrian death: rural/urban environment designation, lighting conditions, driver visibility, driver age, pedestrian age, and precrash vehicle movement.

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1 Introduction

Across the United States of America, traffic deaths have increased in the past decade. Although the overall increase in fatalities is around 22%, pedestrian deaths have increased 50%. The United States is an outlier in this regard; most other high-income countries have experienced a median relative decline in pedestrian fatality rates (Naumann et al., 2025). Thus, identifying the causes of pedestrian fatalities among all traffic deaths is an imperative first step in understanding the changes in causes that underlie this trend.

To identify the causes of pedestrian deaths, we ask: among fatal accidents, what attributes of an accident are causally linked to pedestrian death? To answer this question, we apply modern causal discovery techniques to fatal accident data from the United States. Our methods are applied to more variables and on more data than many other papers, and they support complex claims about causal relationships which are not seen in other parts of the literature.

Background knowledge supports many of our model's claims, and our paper replicates findings related to five key outcome linked variables. Rural/Urban environment designation, lighting conditions, driver visibility, driver age, age of persons involved in the crash, and precrash vehicle movement have all been identified as probable causes of pedestrian death. Our more novel findings include weather's mediation through lighting and the interrelation of road structure level variables. We also find interesting meditations from the age of driver and pedestrian to pedestrian fatalities that merit further investigation.

This paper is structured as follows. First, we discuss the background of our paper providing a literature review, a summary of our data, and an overview of our imputation methods. Next, we discuss our causal identification methods and algorithm. We then discuss the implementation and results of these methods. Finally, we summarize our findings.

2 Background

2.1 Literature Review

A large body of work across multiple disciplines investigates traffic safety broadly and pedestrian deaths specifically. This work has identified a variety of attributes that can influence the severity of pedestrian injuries. The literature breaks these attributes down into five common categories related to the crash. These categories are related to pedestrians, drivers, motor vehicles, roadways, and the environment.

Characteristics across each of these categories have been investigated as possible causes of pedestrian death. The age and sex of drivers and pedestrians have been investigated by Lalika et al., 2022 among other characteristics of road design. Road design, congestion levels, weather, drunkenness, visibility conditions, season, time of week, and day's relationship to the severity of injury in accidents have been investigated by Wang and Wang, 2025 and Yan et al., 2020 among others. The variables the aforementioned authors choose to investigate are impacted by data selection methods and data availability. Wang and Wang, 2025 uses all available data, while Yan et al., 2020 applied fast correlation-based feature selection to narrow the number of features they investigated. Lalika et al., 2022 applied background knowledge and a Bayesian Logistic Regression Model to select variables. All of these authors applied some level of observational selection and coarsening to their features, along with their method of feature selection.

Authors in the field of traffic safety have applied all manner of statistical tools to investigate crash frequency and severity (Lord and Mannering, 2010). Problems with estimation in crash analysis and various model based solutions have been investigated over the years. More recent work has begun to integrate causal discovery work into models as an important step prior to estimation (Lalika et al., 2022, Jin et al., 2025, Wang and Wang, 2025). Researchers use a variety of custom machine learning algorithms and more traditional causal methods to inform their estimation. These methods are also being applied to more granular aspects of traffic safety, such as tunnel accidents and pedestrian subcategories.

2.2 Data

Our data is sourced from the National Highway Traffic Safety Administration's (NHTSA) Fatality Analysis Reporting System (FARS). This system has recorded details on fatal traffic accidents in the United States since 1975 (U.S. Department of Transportation et al., 2024). The data we use are from 2012 to 2024. Our data is at the person level, meaning that each person involved in an accident has their own entry. Accident level data are repeated for each person in the same accident. Vehicle level data had to be constrained to avoid creating sparse variables. We only include vehicle level data for the first vehicle to act in the crash sequence of events. This will attribute vehicle level factors to the vehicle that initiated the crash, not always the one that killed or injured the pedestrian(s).

The FARS variables used in this paper are a subset of the more than 140 unique variables recorded. To select our subset, we applied background knowledge gathered from previous research to sect a list of 33 variables possibly causally related to pedestrian deaths. We transformed some of these into composite variables, such as if a pedestrian died (see appendix A). From there we applied a variety of data quality criteria to further narrow down our list (see appendix B). These criteria were derived from the requirements of our imputation and causal inference algorithms. Reporting sparsity and cardinality were our main concerns.

After selecting the variables of interest, we coarsened the remaining variables with more than 20 categories and binned the discrete variables. Of the remaining variables, two were perfectly correlated with our outcome and had to be dropped. Our resulting data set contained 17 categorical variables (see appendix A).

Our final data set consisted of approximately 1.1 million observations. We used 2023 data to test our code throughout the data pipeline, but did not rely on those tests alone to make decisions about our variables and methods. The number of observations caused us some computational issues. To resolve these issues, we performed random sampling without replacement from the whole data set. We selected our sample sizes for computational reasons. We ran our methods on two subsets of the data, one with 100,000 observations and the other with 250,000. Increasing the sample size changed our outcome as discussed in section 4.

2.3 Variable Imputation

We pursued imputation for two statistical reasons and one practical reason. First, missingness presents an issue of causal interpretation. If we draw an arrow from a binary variable with missing data, it is computed as three categories; we cannot say that the binary variable was the cause if some third category exists. Second, we suspect that our missing data is not drawn from the same distribution as the observed data. Certain kinds of accidents might yield certain types of data. For example, police may be more likely to administer drug tests to certain demographic profiles than others. Dropping the data would shift our distribution and could alter our causal estimations. Based on our concerns, we examined our data to assess imputation as our solution.

Many of our variables contained missing values. The rate of missingness varied (see appendix C). However, the missingness rates were all low enough to meet our heuristic threshold to apply imputation. Our data set was also large and had many samples to base our imputations. Based on our concerns and the suitability of the data, we selected an imputation method.

To perform imputation, we selected the 'missRanger' R package for its computational efficiency and applicability to categorical variables. It is an implementation of the famous 'MissForest' algorithm (Wright and Ziegler, 2017). The imputation was performed on the entire 2012-2024 data set.

3 Methods

3.1 Algorithm

The algorithm we considered to answer our research question is the Really Fast Causal Inference (RFCI) causal discovery algorithm developed in Colombo et al., 2012. The RFCI algorithm is a constraint-based causal discovery algorithm developed as a modification of the Fast Causal Inference (FCI) algorithm posed by Spirtes et al., 2000. The algorithm uses conditional independence oracle information among all variables in a given data set to formulate a Partial Ancestral Graph (PAG) from an initial complete graph among all nodes representing the variables. The algorithm consists of three main steps or "sub-

algorithms" which are discussed at a high level here; more detailed information is available in Appendix D.

The first step of RFCI is to obtain the initial skeleton for the final PAG that will be returned, and it is created from a complete graph among all variables in the given data set. To obtain the initial skeleton, the algorithm uses conditional independence oracle information as an argument to remove edges from the complete graph and obtain a list of all unshielded triples[1].

In the next step of RFCI, we orient the list of unshielded triples from step 1 as v-structures, again according to the conditional independence oracle[2]. The skeleton of the graph obtained from Step 1 is updated to include these now oriented v-structures as well as additionally removed edges determined in Step 2.

In the third and final step, the RFCI algorithm applies a series of ten orientation rules established by Zhang, 2008 to orient remaining edges that have not yet been oriented. RFCI does have one unique modification to one of those rules, the discriminating path rule discussed in Colombo et al., 2012, increasing computational efficiency.

The first main advantage of this algorithm is its greater computational efficiency compared to FCI. This stems from the choice to not consider Possible D-SEP sets delineated in Colombo et al., 2012. Another advantage of RFCI is the ability for it to be deployed in systems with latent and selection variables. This is important in our project since only certain selected variables in the FARS data set are considered for the algorithm, so it is likely some latent variables will emerge. Additionally, as will be discussed in 3.2, additional edge orientations can be applied to the returned PAG from RFCI in a post-hoc context.

Despite the greater computational speed relative to FCI, the iterative nature and incrementing variable set sizes means RFCI algorithm can still be extremely expensive to run. Furthermore, the final PAG returned from RFCI can ultimately be largely uninformative and difficult to interpret, especially in settings with a high number of variables and where background knowledge about edge orientations and independence are not employed.

[1] A set of three vertices X_i, X_j, X_k is called an unshielded triple if X_i and X_j are adjacent, X_j and X_k are adjacent, but X_i and X_k are not adjacent.

[2] An unshielded triple $\langle X_i, X_j, X_k \rangle$ is called a v-structure if X_j is a collider on the path $\langle X_i, X_j, X_k \rangle$.

3.2 Orientation Enforcement

The `rfci()` function in the 'CRAN' package `pcalg` does not provide a built-in interface for incorporating background causal knowledge. Our workaround was to enforce tier constraints post-hoc—after inferring the PAG via `rfci()`—by directly modifying the `@amat` adjacency matrix of the resulting `fciAlgo` object, a concept adapted from STAT 566 Lecture 9 notes (Perković, 2026). The `@amat` matrix encodes PAG edge marks as integers: `0` (no edge), `1` (circle endpoint), `2` (arrowhead), and `3` (tail), where entry `amat[i, j]` records the mark at vertex j on the edge i–j. We enforced temporal directionality by iterating over all variable pairs: for any existing edge spanning variables in different tiers, we set `amat[i, j] = 2` and `amat[j, i] = 3` whenever tier$(i) <$ tier(j), encoding the directed edge $i \to j$ in accordance with our partial temporal causal assumptions.

Partially temporal causal assumptions (Perković, 2026) can be utilized to forbid, force, or directionally orient edges between two nodes. For example, it is reasonable that an environmental variable such as `WEATHER` cannot causally affect a fixed characteristic such as `DR_AGE`, and vice versa — so we forbid this edge entirely. Alternatively, `ROUTE` cannot causally affect `DR_AGE`, but `DR_AGE` plausibly affects `ROUTE` through driver preferences or demographics, so we orient the edge from `DR_AGE` to `ROUTE`. Exhaustively inspecting all $18^2 = 324$ pairwise relationships by hand is not feasible, so we instead group variables into partially temporal tiers that automatically forbid any edge directed from a later tier to an earlier tier.

We define five tiers based on causal priority. Tier 1 contains fixed pre-existing characteristics of the pedestrian and driver: `AGE`, `DR_AGE`, and `DR_SEX`. Tier 2 contains environmental and infrastructure variables that are present prior to any crash event: `DAY_WEEK`, `ROUTE`, `RUR_URB`, `RELJCT1`, `LGT_COND`, `WEATHER`, `VPAVETYP`, `VSURCOND`, and `MECISSUE`. Tier 3 contains pre-crash human state and behavioral variables: `DR_DRINK`, `PREV_ACC`, and `P_CRASH1`. Tier 4 contains crash dynamics variables: `VE_FORMS` and `VIS`. Finally, Tier 5 contains the outcome variable `PED_DEATH`. Variables within a tier may causally affect any variable in a higher-numbered tier, but not vice versa. However, this system is not perfect. Variables within the same tier are not strictly ordered in time relative to one another, so the coarseness of the tiers may mishandle some within-tier and across-tier relationships.

Figure 1 displays the resulting matrix of allowed and forbidden directed edges.



Figure 1: Forbidden directed edges matrix for the 18-variable crash dataset. A red cell at row i, column j indicates that the edge $i \to j$ is forbidden based on the causal tier structure; blue cells indicate allowed edges. Variables are ordered by tier along both axes.

4 Results

4.1 PAG

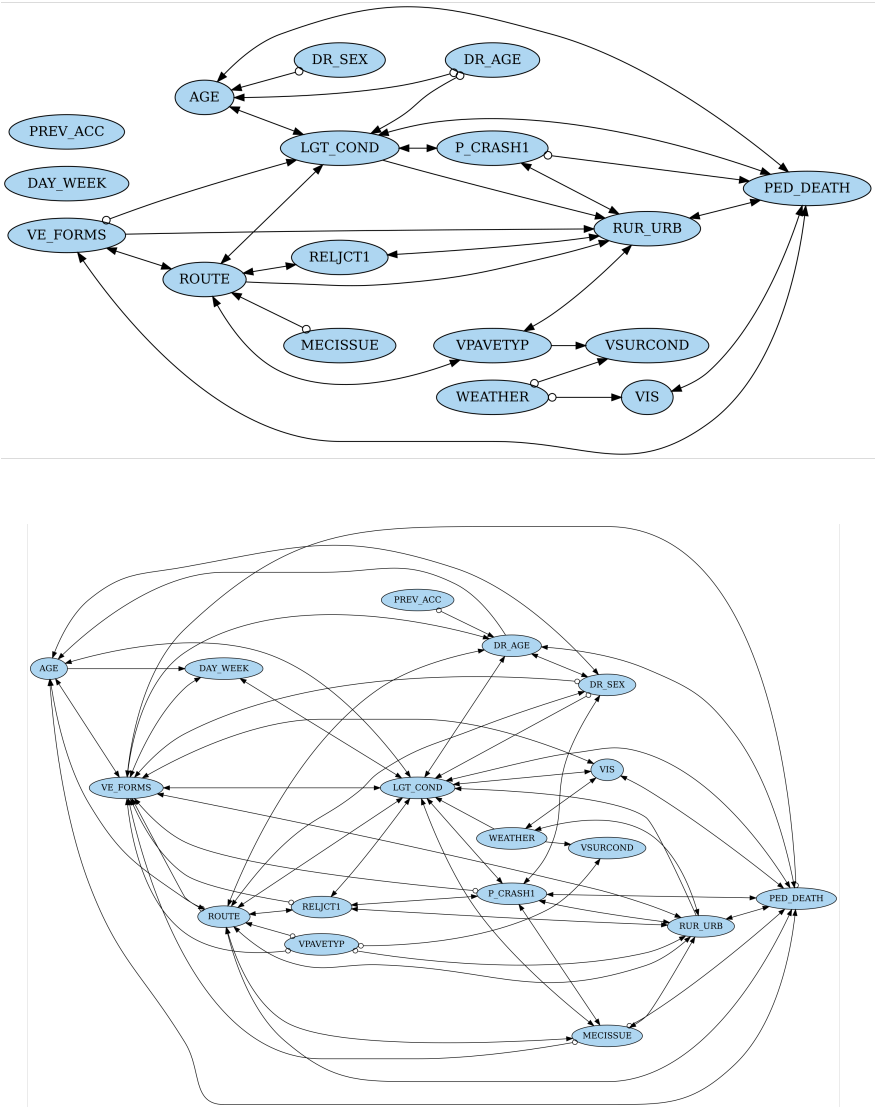


Figure 2: 100 and 250 Thousand Observation Graphs without Enforcement

4.2 DAG

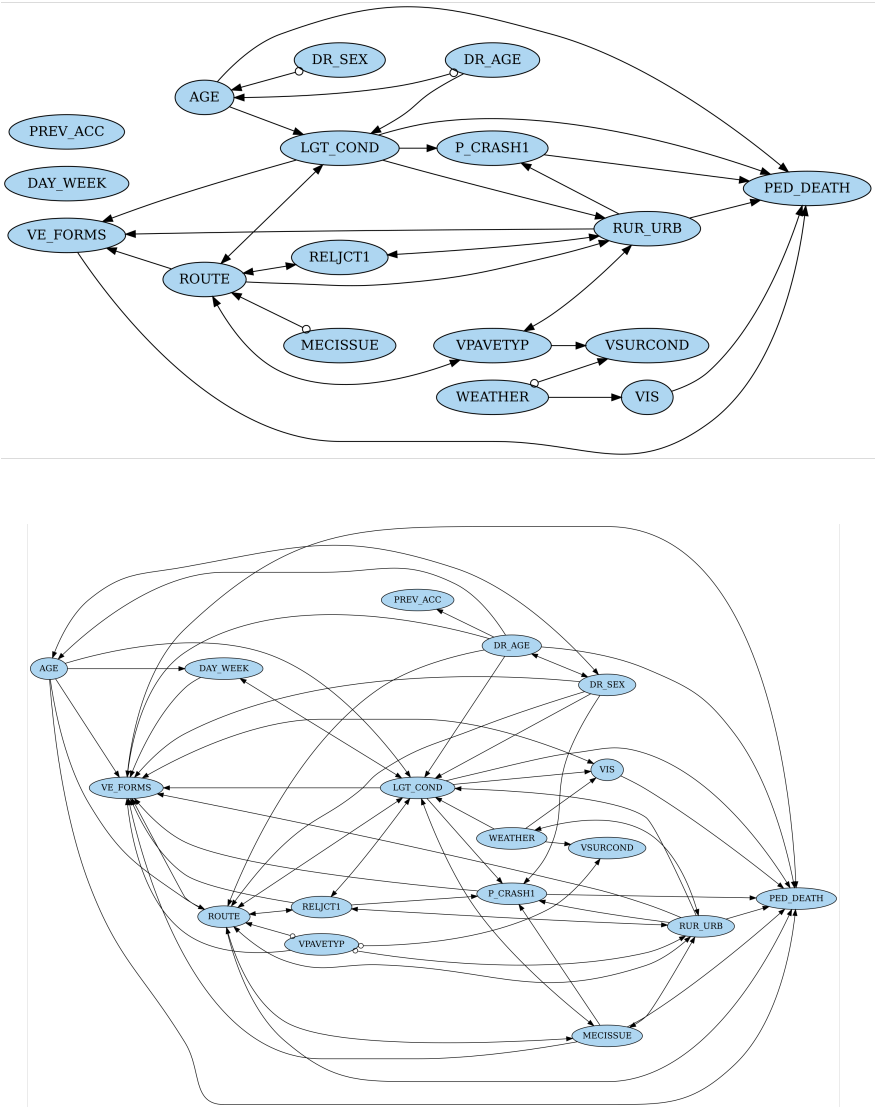


Figure 3: 100 and 250 Thousand Observation Graphs with Enforcement

4.3 Analysis

We inferred two separate PAGs using randomly sampled observations of size 100,000 and 250,000 from an aggregated dataset containing 1.1 million observations. Each PAG was then enforced using the partial temporal causal assumptions. Although they are still incomplete, they are much closer to DAGs, and we will interpret them as DAGs.

The tier-enforced PAG estimated on 250,000 observations reveals a dense but interpretable structure. LGT_COND and VE_FORMS emerge as the two central hubs of the graph, each receiving directed edges from multiple Tier 1 and Tier 2 variables and propagating influence downstream toward PED_DEATH.

AGE and DR_AGE are correlated in all of our DAGS because of a data structure choice. Since every participant in an accident is an observation in the data set drivers are listed as individual observations. The age of the observed driver is always perfectly correlated to the DR_AGE. Since over 50% of our data is the initiating driver in the accident this will cause a strong association in any model.

PED_DEATH receives direct edges from RUR_URB, P_CRASH1, LGHT_COND, VIS, DR_AGE, and AGE, identifying these as the estimated direct causes of pedestrian fatality conditional on all other variables in the graph. RUR_URB itself is a downstream mediator, receiving edges from several environmental and infrastructure variables before directing into the outcome. The path through WEATHER $\rightarrow$ VIS $\rightarrow$ PED_DEATH is also notable, suggesting that adverse weather increases fatality risk primarily through its effect on obstructions of visibility on the driver rather than directly.

VPAVETYP retains a circle endpoint on its edge into ROUTE, indicating that the algorithm could not fully resolve this edge mark from the data alone. This likely reflects a latent common cause between pavement type and route classification rather than genuine bidirectionality, as both variables plausibly share unmeasured infrastructure-level determinants.

PREV_ACC is largely disconnected from the rest of the graph, carrying no edges into PED_DEATH directly or through major mediators. This suggests that prior accident history does not carry detectable conditional associations with pedestrian fatality once road environment and crash dynamics variables are accounted for.

Several edges in the estimated graph are difficult to reconcile with domain knowledge and warrant caution in interpretation. Notably, DR_AGE carries a directed edge into LGT_COND, and both AGE and LGT_COND carry directed edges into DAY_WEEK. None of these relationships are causally plausible: a driver's age cannot determine ambient lighting conditions, and neither age nor lighting can cause the day of the week. These were assumptions not accounted for, a direct limitation of the tier structure discussed earlier.

One important caveat applies to all findings: the FARS data are conditioned on crash

involvement, as only fatal crashes are observed. This selection condition can induce spurious associations between otherwise unrelated variables—a collider-like effect that may manifest as edges in the estimated graph that do not reflect genuine population-level causal relationships.

5 Conclusion

The disproportionate increase in pedestrian fatalities in the United States is a concerning trend. Pedestrians are more likely than motor vehicles to sustain severe injuries in vehicle pedestrian crashes. If we want to live in safe cities, preserve our mobility, and encourage walkability, we need to address the changing risks taken on by pedestrians. Pedestrians cannot be shielded on the road, so the only way to keep them safe is by addressing factors that endanger them before a crash.

This motivation, as indicated above, prompts us to ask the following question: ”Among fatal accidents in the U.S., what causes pedestrian fatalities?” To answer that question we obtained a large set of fatal crash data. We used background knowledge and data constraints to select relevant variables for analysis. Our causal question was confused by missingness among this data. Suspecting distributional differences in our missing data, we used random forest methods and the distribution of known variables among the missing data to impute their categorical values. This transformed data was suited to our causal estimation methods and allowed us to apply RFCI to investigate our question.

Our application of RFCI to the data generated a useful PAG preserving the potential effects of latent variables with directional uncertainty and unknown edge structures. We applied background knowledge in a tiered structure to move towards a DAG and use temporal relationships to establish the directional relationships the PAG could not capture. For the data with 100 thousand observations compared to that with 250 thousand observations, the PAG and DAG maintained many of the same edges and structures. The model became more confident in asserting directed edges in several places, and it included two variables that had no edges before. The many structural similarities of these two models gives us confidence in the underlying skeleton of our model.

The model provided interesting insights compatible with our background knowledge that suggest interesting relationships between factors. The model suggests lighting conditions

and visibility mediate weather effects on pedestrian death. Lighting condition was also one of our densest nodes, suggesting it mediates many other effects on pedestrian death. The model also suggested some spurious relationships based on our background knowledge. It is unlikely that age causes lighting conditions or the day of the week. Our model mediates age's effects on pedestrian fatality through day of the week, lighting conditions, and route among others. These relationships might be explained by a behavioral mediator that explains how retired people and children's lifestyles differ from that of working age people. These differences merit further investigation.

Two major limiting factors affect our work. First, data limitations, specifically sparsity and size, affected our ability to include variables and to utilize all of our data in a reasonable time frame. If RFCI could be parallelized or if we could find a more computationally efficient method, it would allow us to use more of our data and perhaps draw stronger conclusions. Furthermore, more complete data across categories would enable a more throughout investigation and reduce our reliance on imputation to draw causal conclusions.

Second, our methods rely on an oracle test for conditional independence that must be performed hundreds of times to compute all relationships between our data. This issue is an inherent attribute of using constraint-based causal discovery methods, but nonetheless, it affects our confidence in our findings. With this many variables over 250 independence tests were performed on the same data. A method that takes into account that repeated testing or a score based method that can take into account latent variables might provide a more statistically sound estimation. Developing these methods would be important to far more than traffic safety analysis, and it remains an ongoing area of research in causal inference.

A Data Dictionary

National Transportation Atlas Database (NTAD): Fatality Analysis Reporting System full data dictionary: National Center for Statistics and Analysis, 2026

For full details on each of these variables and their history refer to the dictionary. Variables in our model not listed below were not changed before being used in our model.

A.1 NMCRASH Transformation

The variable NMCC has multiple instances per non-moterist person in the data set. These unique codes are simplified from approximately 90 unique variable combinations into 5 simple categories based on similarity in categorization and possibly unique causal factors. Several possibly unique causal effects had to be merged into others.

`NMCCSimple`

Value	Description
0	None, (NMCC has value 0)
1	NMCC 2
2	Some combo of in the road improperly (NMCC 3, 4, 11, 12)
3	Visibility Issue of NM (contains NMCC 19)
4	Other (Infrequently used causes)
5	Unknown (NMCC has value 98, 99)

A.2 Factor Transformation

The FACTOR data set indicates whether a vehicle had a known mechanical issue prior to the crash. It also specifies the kind of mechanical issue. This data set is incredibly sparse with 98+% of the data being none/unknown. It was simplified into three categories to deal with this sparsity.

MECISSUE

Value	Description
0	None (0)
1	Mechanical Issue, (All other values)
2	Unknown (98 or 99)

A.3 Distracted Transformation

The DISTRACT data set contains information on if the driver was distracted and if so what by. It is also relatively sparse and therefore is aggregated into distracted or not. A larger proportion of this data was reported as none. Since the driver being missing is a potential unique causal effect unrelated to distraction it was left as separate.

DISTRACT

Value	Description
0	None (0)
1	Distracted, (All other values)
2	Driver not present/unknown if present (16)
3	Unknown/Not Reported (96 or 99)

A.4 Vision Transformation

This is a binary variable predicting only ˜7% of cases have a noted visual impairment. Rain, Snow, Fog, Smoke, Sand, Dust is the most commonly cited visual impairment leaving very few other instances. This was aggregated into an essentially binary variable. Since the driver being missing is a potential unique causal effect unrelated to visual impairments it was left as separate.

VIS

Value	Description
0	None (0)
1	Vision Obscured, (All other values)
2	Driver not present/unknown if present (95)
3	Unknown/Not Reported (99)

A.5 Pedestrian Death

Binary Variable for pedestrian death. This is the paper's outcome variable of interest. It aggregates pedestrians and "other pedestrians" from the FARS data. This means it includes people killed in cherry pickers and in buildings among other uncommon circumstances.

PED_DEATH

Value	Description
0	Pedestrian didn't Died
1	Pedestrian died

A.6 Driver Age and Age

Age of the driver of the crash initiating vehicle or age of any person involved in the accident.

DR_AGE

Value	Description
0-15	Child
16-25	Young Adult
26-64	Adult
65+	Elderly

A.7 Driver Sex

Age of the driver of the crash initiating vehicle.

DR_SEX

Value	Description
1	Male
2	Female
8	Unreported
9	Unknown

A.8 Driver Drugs

Drug involvement of the driver of the crash initiating vehicle.

DR_DRUGS

Value	Description
0	No (Drugs Not Involved)
1	Yes (Drugs Involved)
8	Not Reported
9	Unknown (Police Reported)

A.9 Emergency Vehicle Arrival and Death

This variable's goal is to categorize at the person level whether the emergency vehicle arrival time matters. It follows this algorithm:

1. If the time of the crash is unknown assign it a value of -1
2. Compare the time of death to the time of the crash, if it is within 5 minutes set to 0
3. If not set to 0 compute the time it took EMS to arrive at the scene and assign the variable that value

It is assumed that EMS always arrived within 24 hours for the sake of computation since no date is listed.

EMSAD

Value	Description
0	Died within 5 minutes of crash
6+	A discreet number of minutes between the crash and EMS arrival
-1	Unknown time of crash

A.10 Pre-Crash Activity

Condenses several categories into broader categories preserving causal relationships as much as possible.

P_CRASH1

Value	Description
straight travel	1, 2, 3
stopped or parking	4, 5, 7, 8, 9
complex maneuver	6, 10–17, 98
NA	0, 99 (Unknown/Missing)

A.11 Number of Vehicles

Number of vehicles involved in an given accident. This is a discreet variable and is binned based on frequency. Most accidents in the data set involve only one or two variables.

VE_FORMS

Value	Description
single	1
two	2
multiple	3 or more
NA	Unknown/Missing

A.12 Route Signage

Kind of road accident occurred on. This variable is coarsened into several higher level categories attempting to preserve causal relationships.

ROUTE

Value	Description
interstate	1, 11
highway	2, 3
local	4, 5, 6, 12 (county/township/municipal)
unsigned	0, 96 (not signed + not in inventory)
other	10, 13, 95
NA	99 (Unknown/Missing)

A.13 Weather Conditions

Infrequent cold weather and other effects condensed into single variables. This merges highly infrequent variables.

WEATHER

Value	Description
clear	1
cloudy	10
rain	2
frozen_precip	3, 4, 11, 12 (snow/sleet/freezing rain)
fog	5
other	6, 7, 8
NA	98, 99 (Unknown/Missing)

A.14 Previous Accidents

Discreet variable binned. More then 1 previous accident is highly infrequent in the data.

PREV_ACC

Value	Description
none	0
one	1
multiple	2 or more
NA	98, 99, 998 (Unknown/Missing)

A.15 Pavement Type

Coarsened road types into paved and unpaved to condense infrequent variables.

VPAVETYP

Value	Description
paved	1, 2, 3 (concrete/asphalt/brick)
unpaved	4, 5 (gravel/dirt)
non_trafficway	0
other	7
NA	8, 9 (Unknown/Missing)

A.16 Surface Condition

Road surface conditions. Coarsening infrequent variables and condensing cold weather related effects to merge infrequently occurring categories.

VSURCOND

Value	Description
dry	1
wet	2
frozen	3, 4, 10 (snow/ice/slush)
other	0, 5, 6, 7, 8, 11 (sand/water/oil/mud etc.)
NA	98, 99 (Unknown/Missing)

A.17 Light Condition

Condensed infrequently used codes with similar meanings.

LGT_COND

Value	Description
daylight	1
dark_unlit	2
dark_lighted	3
dawn_dusk	4, 5
dark_unknown	6, 7
NA	8, 9 (Unknown/Missing)

B Variable Selection Processes

Table 1 shows the variables that made it through each stage of our section possess. Since the total FARS variable list is prohibitively long, we begin with only variables selected based on a generous interpretation of background knowledge (see Macioszek et al., 2023, Wang and Wang, 2025, Yan et al., 2020, Lord and Mannering, 2010). Several of the variables below are our transformations of other variables(see appendix A). Our second pass identified missingness beyond what was safe to impute (see appendix C). The final step was to remove variables perfectly correlated with our outcome since this confuses our causal inference models (this is an open area of study).

Variable	Background Knowledge	Missingness	Perfect Correlation
VSPD_LIM	Y	N	N
TRAV_SP	Y	N	N
DRINKING	Y	N	N
DR_DISTRACT	Y	N	N
DRUGS	Y	N	N
DR_DRUGS	Y	N	N
EMSAD	Y	N	N
NMCC	Y	Y	N
M_HARM	Y	Y	N
PED_DEATH	Y	Y	Y
VE_FORMS	Y	Y	Y
ROUTE	Y	Y	Y
RUR_URB	Y	Y	Y
LGT_COND	Y	Y	Y
WEATHER	Y	Y	Y
DR_DRINK	Y	Y	Y
PREV_ACC	Y	Y	Y
VSURCOND	Y	Y	Y
P_CRASH1	Y	Y	Y
AGE	Y	Y	Y
DR_Age	Y	Y	Y
DR_SEX	Y	Y	Y
MECISSUE	Y	Y	Y
DAY_WEEK	Y	Y	Y
RELJCT1	Y	Y	Y
VPAVETYP	Y	Y	Y

Table 1: Variable Selection Steps

C Data Summary Statistics

C.1 Missingness

Variable	Missing (%)
VPAVETYP	25.2
PREV_ACC	11.2
MECISSUE	7.4
ROUTE	7.1
WEATHER	3.8
VIS	3.7
DR_AGE	3.1
DR_SEX	2.9
AGE	2.4
P_CRASH1	1.2
VSURCOND	1.1
LGT_COND	0.4
RUR_URB	0.1
RELJCT1	0.1
VE_FORMS	0.0
DAY_WEEK	0.0
DR_DRINK	0.0
PED_DEATH	0.0

Table 2: Missingness Rates by Variable

Our heuristic was around 10% missingness was justified to be imputed. One variable exceeded this threshold, but we made an exception because the variable was deemed important.

C.2 Variable Frequency

The following are tables displaying the categorical frequency across all 17 variables in our data. Tables are all in percentages.

	Adult	Child	Elderly	Yong Adult
Age	57.6	8.0	13.0	21.4
Driver Age	63.4	0.5	11.8	24.3

Table 3: Frequencies: AGE

	Multiple	Single	Two
Number of Vehicles	15.1	42.3	42.7

Table 4: Frequencies: VE_FORMS

	1	2	3	4	5	6	7
Weekday	15.8	12.7	12.2	12.5	13.2	15.8	17.9

Table 5: Frequencies: DAY_WEEK

	Highway	Interstate	Local	Other	Unsigned
Road Type	51.3	14.4	31.3	1.2	1.8

Table 6: Frequencies: ROUTE

	No	Yes
Relationto Junction	94.7	5.3

Table 7: Frequencies: RELJCT1

	Dark Lighted Lighted	Dark Unknown	Dark Unlit	Dawn Dusk	Daylight
Lighting Condition	20.4	0.8	25.7	4.2	48.9

Table 8: Frequencies: LGT_COND

	Clear	Cloudy	Fog	Frozen/Precip	Other	Rain
Weather	75.3	14.2	1.3	1.6	0.3	7.3

Table 9: Frequencies: WEATHER

	Multiple	None	One
Previous Accident	3.8	84.0	12.2

Table 10: Frequencies: PREV_ACC

	Non-trafficway	Other	Paved	Unpaved
Road Surface	0.5	0.1	98.2	1.2

Table 11: Frequencies: VPAVETYP

	Dry	Frozen	Other	Wet
Surface Conditions	85.6	2.1	1.0	11.3

Table 12: Frequencies: VSURCOND

	Complex Maneuver	Stopped or Parking	Straight Travel
Pre-Crash Move	29.2	0.6	70.2

Table 13: Frequencies: P_CRASH1

	No	Yes
Mechanical Issue	97.2	2.8

Table 14: Frequencies: MECISSUE

	No	Yes
Driver Visibility Issues	96.6	3.4

Table 15: Frequencies: VIS

	Male	Female
Driver Sex	76.1	23.9

Table 16: Frequencies: DR_SEX

	Yes	No	Unknown
Rural/Urban	42.6	57.2	0.2

Table 17: Frequencies: RUR_URB

	No	Yes
Pedestrian Death	92.5	7.5

Table 18: Frequencies: PED_DEATH

D Full Algorithm Summary

The Really Fast Causal Inference (RFCI) algorithm takes the conditional independence oracle information about a complete undirected graph $\mathcal{C}$ from its node set $\mathbf{X} \cup \mathbf{S}$, and it will return a Partial Ancestral Graph (PAG). Refer to Colombo et al., 2012 for the original development and theory of the algorithm in detail.

The following three sub-algorithms make up the overall RFCI algorithm:

- **Sub-algorithm 1 (S1)**: From the complete graph $\mathcal{C}$, we return the initial skeleton for the PAG , separation sets (sepset), and an unshielded triple list using conditional independences $\mathcal{M}$.
- **Sub-algorithm 2 (S2)**: With the initial skeleton for the PAG ($\mathcal{C}$), separation sets (sepset), and an unshielded triple list $\mathcal{M}$ from S1, we then orient the v-structures for the PAG and update the skeleton and sepset as well. The algorithm returns the partially oriented graph ($\mathcal{C}$) and separation sets (sets).
- **Sub-algorithm 3 (S3)**: With the partially oriented graph and separation set retrieved from S2, we then orient the remaining edges according to a list of rules.

D.1 Step 1

This algorithm takes the conditional independence (oracle) information among all selected variables in the vertex set for a complete undirected graph $\mathcal{C} = (\mathbf{X}, \mathbf{E})$. Let $\mathbf{S}$ denote the selection variables. The goal of this sub-algorithm is to retrieve the initial skeleton of the ultimate PAG, as well as obtain the v-structures and separation sets.

Note the following definitions:

- sepset(X_i, X_j): If there exists a set $\mathbf{Y}$ that makes X_i and X_j conditionally independent given $\mathbf{Y}$, the edge between X_i and X_j is removed, then the set $\mathbf{Y}$ is the separation set and is saved in sepset$(X_i, X_j) =$ sepset(X_j, X_i)
- Unshielded triple $\langle X_i, X_j, X_k \rangle$: Three vertices X_i, X_j, X_k are called an unshielded triple if X_i and X_j are adjacent, X_j and X_k are adjacent, but X_i and X_k are not adjacent.

Algorithm 1 Sub-algorithm 1

Require: Conditional independence oracle information among all $\mathbf{X}$ given $\mathbf{S}$, which is the vertex set for the graph $\mathcal{C}$.

Form the complete graph $\mathcal{C}$ on the vertex set $\mathbf{X}$ with the edges $\circ\!\!-\!\!\circ$
$n = -1$
repeat
  $n = n + 1$
  repeat
    for all vertices X_i in C **do**
      Compute adj(C, X_i)
    end for
    Select a new ordered pair of vertices (X_i, X_j) that are adjacent in $\mathcal{C}$ and satisfy |adj$(C, X_i)\backslash\{X_j\}| \geq n$
    while X_i and X_j are adjacent in $\mathcal{C}$ and $\exists\ \mathbf{Y} \subseteq$ adj$(\mathcal{C}, X_i)\backslash\{X_j\}$ with $|\mathbf{Y}| = n$ that has not yet been considered **do**
      Choose a (new) set $\mathbf{Y} \subseteq$ adj$(\mathcal{C}, X_i)\backslash\{X_j\}$ with $|\mathbf{Y}| = n$
      if X_i and X_j are conditionally independent given $\mathbf{Y} \cup \mathbf{S}$ **then**
        Delete edge $\circ\!\!-\!\!\circ$ from $\mathcal{C}$
        Let sepset(X_i, X_j) = sepset$(X_j, X_i) = \mathbf{Y}$
      end if
    end while
  until all ordered pairs of adjacent vertices (X_i, X_j) in $\mathcal{C}$ with |adj$(C, X_i)\backslash\{X_j\}| \geq n$ have been considered
until all pairs of adjacent vertices (X_i, X_j) in $\mathcal{C}$ satisfy |adj$(C, X_i)\backslash\{X_j\}| \leq n$
Form a list $\mathcal{M}$ of all unshielded triples $\langle X_k, \cdot, X_m \rangle$ in C with $k < m$
return $\mathcal{C},$ sepset, $\mathcal{M}$

D.2 Step 2

This algorithm takes the three items returned from **S1**–$\mathcal{C},$ sepset, and $\mathcal{M}$–and uses them to orient v-structures in our graph skeletion $\mathcal{C}$ The goal of this algorithm is to update $\mathcal{C}$ to be a partially oriented graph with these v-structures oriented and separation sets. The partially oriented graph $\mathcal{C}$ and the sepset are returned.

Note the following definition:

- v-structure: An unshielded triple $\langle X_i, X_j, X_k \rangle$ is called a v-structure if X_j is a collider on the path $\langle X_i, X_j, X_k \rangle$.

Algorithm 2 Sub-algorithm 2

Require: Initial skeleton $\mathcal{C}$, separation sets (sepset), and unshielded triple list $\mathcal{M}$ (from S1).

Let $\mathcal{L}$ be an empty list.
while $\mathcal{M}$ is non-empty **do**
    Choose an unshielded triple list $\langle X_i, X_j, X_k \rangle$ from $\mathcal{M}$
    if both $(X_i$ and $X_j)$ and $(X_j$ and $X_k)$ are conditionally dependent given sepset$(X_i, X_k) \backslash \{X_j\}$ **then**
        Add $\langle X_i, X_j, X_k \rangle$ to $\mathcal{L}$
    else
        for $r \in \{i, k\}$ **do**
            if X_r and X_j are conditionally independent given sepset$(X_i, X_k) \backslash \{X_j\}$ **then**
                Find a minimal separating set $\mathbf{Y} \subseteq$ sepset$(X_i, X_k) \backslash \{X_k\}$ for X_r and X_j
                Let sepset$(X_r, X_j) =$ sepset$(X_j, X_r) = \mathbf{Y}$
                Add to $\mathcal{M}$ and $\mathcal{L}$ all triples $\langle X_{\min(r,j)}, \cdot, X_{\max(r,j)} \rangle$ that form a triangle in $\mathcal{C}$
                Delete from $\mathcal{M}$ and $\mathcal{L}$ all triples containing $\langle X_r, X_j \rangle$ of the form: $\langle X_r, X_j, \cdot \rangle, \langle X_j, X_r, \cdot \rangle, \langle \cdot, X_r, X_j \rangle$, and $\langle \cdot, X_j, X_r \rangle$
                Delete edge $X_r * - * X_j$ from $\mathcal{C}$
            end if
        end for
    end if
    Remove $\langle X_i, X_j, X_k \rangle$ from $\mathcal{M}$
end while
for all elements $\langle X_i, X_j, X_k \rangle$ of $\mathcal{L}$ **do**
    if $X_j \notin$ sepset(X_i, X_k) and both edges $X_i * - * X_j$ and $X_j * - * X_k$ are present in $\mathcal{C}$ **then**
        Orient $X_i * - \circ X_j \circ - * X_k$ as $X_i * \rightarrow X_j \leftarrow * X_k$
    end if
end for
return Partially Oriented Graph $\mathcal{C}$, sepset

D.3 Step 3

This algorithm takes the arguments of $\mathcal{C}$, (the partially oriented graph from **S2**) and the separation sets (sepset) as well. With this algorithm, we ultimately orient the remaining edges in the graph according to the set of rules established in Zhang, 2008. Additionally, one unique modification to the discriminating path rule is made as discussed in Colombo et al., 2012, and it is detailed in the pseudo-code for the algorithm in **S3**. With these orientations we obtain the Partial Ancestral Graph $\mathcal{C}$ and the sepset, as these are what is returned by the algorithm.

Note the following definitions taken from Zhang, 2008:

- Discriminating Path p: In a Maximal Ancestral Graph, a path between X and Y, $p = \langle X, ..., W, V, Y \rangle$ is a discriminating path for V if:
  (i) p includes at least three edges;
  (ii) V is a non-endpoint vertex on p, and is adjacent to Y on p; and
  (iii) X is not adjacent to Y, and every vertex between X and V is a collider on p and is a parent of Y.
- Uncovered Path: In a Partial Mixed Graph, a path $p = \langle V_0, \ldots, V_n \rangle$ is said to be uncovered if for every $1 \leq i \leq n-1$, V_{i-1} and V_{i+1} are not adjacent, i.e., if every consecutive triple on the path is unshielded.
- Potentially Directed: In a Partial Mixed Graph, a path $p = \langle V_0, \ldots, Vn \rangle$ is said to be potentially directed from V_0 to V_n if for every $0 \leq i \leq n-1$, the edge between V_i and V_{i+1} is not into V_i or out of V_{i+1}.

The algorithm orients the edges in $\mathcal{C}$ according to the following rules from Zhang, 2008:

1. If $\alpha *\!\!\rightarrow \beta \circ\!\!-\!\!* \gamma$, and α and γ are not adjacent, then orient the triple as $\alpha *\!\!\rightarrow \beta \rightarrow \gamma$
2. If $\alpha \rightarrow \beta *\!\!\rightarrow \gamma$ or $\alpha *\!\!\rightarrow \beta \rightarrow \gamma$, and $\alpha *\!\!-\!\!\circ \gamma$, then orient $\alpha *\!\!-\!\!\circ \gamma$ as $\alpha *\!\!\rightarrow \gamma$
3. If $\alpha *\!\!\rightarrow \beta \leftarrow\!\!* \gamma$, $\alpha *\!\!-\!\!\circ \theta \circ\!\!-\!\!* \gamma$, α and γ are not adjacent, and $\theta *\!\!-\!\!\circ \beta$, then orient $\theta *\!\!-\!\!\circ \beta$ as $\theta *\!\!\rightarrow \beta$
4. For every (remaining) $\alpha \circ\!\!-\!\!\circ \beta$, if there is an uncovered circle path $p = \langle \alpha, \gamma, \ldots, \theta, \beta \rangle$ between α and β such that $\alpha,\ \theta$ are not adjacent and $\beta,\ \gamma$ are not adjacent, then orient $\alpha \circ\!\!-\!\!\circ \beta$ and every edge on p as undirected edges

5. If α–β○–*γ (α and γ may or may not be adjacent), then orient β○–*γ as β–*γ

6. If α –○β○–*γ, and α, γ are not adjacent, then orient β○–*γ as β-*γ

7. If $\alpha \rightarrow \beta \rightarrow \gamma$ or α –○$\beta \rightarrow \gamma$ and α○→γ, orient α○→γ as $\alpha \rightarrow \gamma$

8. If α○→γ and $p = \langle \alpha, \beta, \theta, \ldots, \gamma \rangle$ is an uncovered potentially directed path from α to γ such that γ and β are not adjacent, then orient α○→γ as $\alpha \rightarrow \gamma$.

9. Suppose α○→γ, $\beta \rightarrow \gamma \leftarrow \theta$, p_1 is an uncovered potentially directed path from α to β, p_2 is an uncovered potentially uncovered path from α to θ. Let μ be the vertex adjacent to α on p_1 (μ can be β), and ω be the vertex adjacent to α on p_2 (ω could be θ). If μ and ω are distinct and not adjacent, then orient α○→γ as $\alpha \rightarrow \gamma$.

Algorithm 3 Sub-algorithm 3

Require: Partially oriented graph ($\mathcal{C}$) and separation sets from **S2**

repeat
  Orient as many edge marks as possible in $\mathcal{C}$ by applying rules 1-3 above
  while a triangle between three vertices $\langle X_l, X_j, X_k \rangle$ exists such that $X_j \circ\!\!-\!\!* X_k$, $X_l \leftarrow\!\!* X_j$, and $X_l \rightarrow X_k$ in $\mathcal{C}$ **do**
    Find a shortest discriminating path for $\langle X_l, X_j, X_k \rangle$
    if a discriminating path π for $\langle X_l, X_j, X_k \rangle$ exists between X_k and say X_i **then**
      for all pairs of vertices X_r, X_q that are adjacent on π **do**
        Let $n = -1$;
        while |sepset$(X_i, X_k) \backslash \{X_r, X_q\}| \geq n$ **do**
          Let $n = n + 1$
          while X_r and X_q are adjacent in $\mathcal{C}$ and $\exists$ $Y \subseteq$ sepset$(X_i, X_k) \backslash \{X_r, X_q\}$ with $|Y| = n$ that has not been considered **do**
            Choose a (new) subset $\mathbf{Y} \subset$ sepset$(X_i, X_k) \backslash \{X_r, X_q\}$ with $|\mathbf{Y}| = n$.
            if X_r and X_q are conditionally independent given $\mathbf{Y} \cup \mathbf{S}$ **then**
              Let sepset$(X_r, X_q) =$ sepset$(X_q, X_r) = \mathbf{Y}$
              Create a list $\mathcal{M}$ of all triples $\langle X_r, \cdot, X_q \rangle$ with $r < q$ that form a triangle in $\mathcal{C}$
              Delete edge $X_r *\!\!-\!\!* X_j$ from $\mathcal{C}$
              Run Algorithm **S2** with input $\{\mathcal{C}$, sepset, $\mathcal{M}\}$ and update $\mathcal{C}$ and sepset.
            end if
          end while
        end while
      end for
      if all of the edges between adjacent vertices on π are present in $\mathcal{C}$ **then**
        if $X_j \in$ sepset(X_i, X_k) **then**
          Orient $X_j \circ\!\!-\!\!* X_k$ as $X_j \rightarrow X_k$ in $\mathcal{C}$
        else
          Orient the triple $X_l \leftarrow\!\!* X_k \circ\!\!-\!\!* X_k$ as $X_l \longleftrightarrow X_j \longleftrightarrow X_k$ in $\mathcal{C}$
        end if
      end if
    end if
  end while
  Orient as many edges as possible in $\mathcal{C}$ by applying rules 4-9 above.
until $\mathcal{C}$ remains unchanged throughout lines 2-30 above
return Partial Ancestral Graph $\mathcal{C}$, sepset

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